

SOLUTION

The Water Jug Problem is solved by exploring different states of the two jugs until the required quantity of water is obtained.

1. Represent the amount of water in each jug as a state.
2. Define the capacities of the two jugs and the target quantity.
3. Start with both jugs empty.
4. Generate all possible valid operations:
 - Fill the first jug.
 - Fill the second jug.
 - Empty the first jug.
 - Empty the second jug.
 - Pour water from the first jug to the second jug.
 - Pour water from the second jug to the first jug.
5. After each operation, generate a new state.
6. Avoid revisiting previously explored states.
7. Continue exploring valid states until the target quantity is obtained in either jug.
8. Display the sequence of operations that leads to the solution.